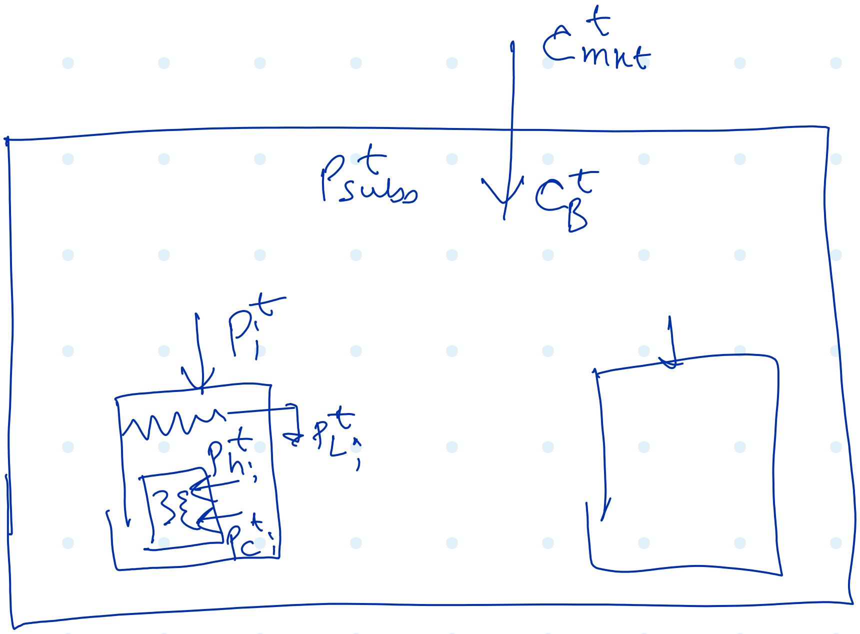

B : Set of Buildings.

Bilinear? (Nonconvex)

needs discretization of C^t_B and big M relaxation

$$\min. \sum_{t=1}^{T} \Big[\underbrace{\left(C^t_{mkt} - C^t_B\right) P^t_{subs}}_{\text{DSO Power Exchange Loss}} + \sum_{i \in B} \left(D^t_i + \alpha SHC^t_i\right) \Big]$$

$$\alpha \left\{ \eta_{h_i} P^t_{h_i} + \frac{1}{\eta_{c_i}} P^t_{c_i} \right\}$$

s.t.

- Network Real Power Balance :=

$$\boxed{P^t_{subs} = \sum_{i \in B} P^t_i \qquad \forall\ t \in T}$$

- Building Real Power Balance:

$$\boxed{P^t_i = \underbrace{P^t_{L_i}}_{\text{Load (param)}} + \underbrace{P^t_{H_i}}_{\text{HVAC}} \qquad \forall\ i \in B\ \forall\ t \in T}$$

$$\boxed{P^t_{H_i} = \underbrace{P^t_{h_i}}_{\text{Heating}} + \underbrace{P^t_{c_i}}_{\text{Cooling}} \quad \forall i \in B \ \forall t \in T}$$

- Building Temperature Trajectory:

$$\boxed{T_i^t = T_i^{t-1} + \Delta t \left(\underbrace{\eta_{h_i} P^t_{h_i}}_{\text{Heating}} - \underbrace{\frac{1}{\eta_{c_i}} P^t_{c_i}}_{\text{Cooling}} - \underbrace{\lambda_i (T_i^{t-1} - T^t_{ext})}_{\text{Decay}} \right) \quad \forall i \in B \ \forall t \in T}$$

- Building Discomfort Penalty Cost.

$$\boxed{D_i^t = \underbrace{K_i}_{\text{Sensitivity}} \times (T_i^t - T_*)^2 \quad \forall i \in B \ \forall t \in T}$$

- Turn off HVAC if energy too costly

$$\boxed{u_i^t \in \{0, 1\}} \quad \boxed{\forall i \in B, t \in T}$$

Hyperparameter $C_{Max} = Max\{\overline{C}_1, \overline{C}_2, \ldots \overline{C}_B\} + 1$

("big-M")

$$\boxed{(c_B^t - c_i) \le (c_{max} - c_i)(1 - u_i^t) \quad \forall i \in B \ \forall t \in T,}$$

$$\boxed{0 \le P_{H_i}^t \le u_i^t \cdot \overline{P}_{H_i}}$$

- Substation Power Constraint:

$$\boxed{P_{sub}^t \ge 0}$$